

The use of integrated circuits by the US military in the 1960s marked the beginning of widespread use of the new technology

Evidence from the VLSI world may suggest that Moore's Law, which states that technical advances relating to chips double every 2 years, is slowing down

country, where companies engage in open forums and interviews. Internship is another means. Intel has a very strong internship programme. Hiring engineering students as interns and having them work with their design teams lets students get a first-hand experience of what it's like to work in this industry. At the end, companies take on the better ones as fulltime employees. This year, Intel has over 100-150 interns working in different development groups across India.

The intake varies widely from company to company. AMD believes in taking only a few people at a time. This is due to the steep learning curve. When it comes to microprocessor design, it doesn't use large teams. The Athlon design was done with less than 200 people. Therefore, finding the right talent is more important than making up the numbers.

Training

Some reports suggest that while sourcing talent, semiconductor companies find that there is a lack of engineers who can immediately be deployed on a design and that Indian electronics engineers need additional training and skill development. However, many industry insiders we spoke to beg to differ. In fact, they believe Indian engineers come with strong conceptual understanding and impressive learning aptitude. They need to be trained only on the latest tools and development methodologies as they are not usually exposed to these at the college level. However, they are able to pick up everything they need pretty quickly.

When working on projects for designing low power chips, efficient multi-core architectures, mixed signal chips, memory sub-systems, interrupt controllers, and interfacing with video/audio processor, the basic textbook knowledge may be insufficient.

"Current curriculum touches the basics of computer architecture, while taking up a school project and learning the relevant design automation tools, from companies like Cadence / Mentor Graphics / Synopsys will be a practical help," said Sandeep.

Some companies have strong mentorship programmes whereby junior engineers are paired with a senior engineer who coaches and guides them along their way. This is of course in addition to regular training programmes.

Work culture

It's not all work and no play when it comes to a career in chip designing. It's a very young industry to work in and the average age of employees at most companies is low, making the work culture active and vibrant. Companies have extremely challenging programmes where the teams are motivated to deliver; but at the same time, there are several recreational activities such as team building, sports and quarterlies. Intel organises events that involve employees as well as their families such as cricket sessions and tennis matches. So it's a mixture of cutting edge work and fun.

Career path

Career paths in this industry are well defined for aspirants in both technical and management functions. Typically, an engineer takes about 5 – 6 years to reach the position of a Senior Design Engineer. As the aspirant gains more work experience by working with different designs, he / she can advance to a Lead position. Post this level, the engineer can



"Seeing technologies you've worked on, being put into day to day use is quite an exhilarating experience!"

Rohit Vidwans
Director,
Digital Enterprise Group,
Intel India.

The right Course

Companies consider Tier A colleges in India such as IITs, BITS, NITs, etc., as a good source of qualified engineers. However, candidates from regional colleges such as VJTI or Manipal Institute are also considered, provided these students can opt for a course in VLSI design. Companies look for students at undergraduate and masters levels in computer science and electronics domains with a consistently high GPA. Here's a list of some of the top places that offer courses in VLSI Design and allied fields, as recommended by VLSI chip design companies.

CDAC, Hyderabad:

Certificate Course in Digital and Analog VLSI Design (CDAD)

Sandeepani School of VLSI Design, Bangalore & Hyderabad:

Post-graduate diploma in VLSI design

Veda IIT, Hyderabad:

MS and Diploma in VLSI and Embedded systems

Cranes Software, Bangalore:

Embedded System certification

Vedant Chandigarh, Lucknow: Course in front-end and back-end design including Analogue & Mixed Signal design

decide if he / she wants to continue in the technical function or move to a management function such as Project Manager, both of which are equally fulfilling depending on the inclination. How quickly you grow depends entirely on the individual ability. You can also branch off and choose to become an expert in a certain technology, for instance, L2 design, and be a mentor or Senior Fellow.

As for the salaries in this industry, a fresher starts off at around 5 lakhs per annum while a Lead Engineer generally gets 8 – 12 lakhs. In a few years, one can progress to Project Lead where 12 – 20 lakhs is the norm and the sky is the limit, depending on the brand.

Summing it up

To make a successful career one does need to have a special interest in the subject and a dream to make a difference. Conscious choices have to be made along the way which may be less rewarding in the short run such as going in for a focused Masters with a good project. The money and lucrative job offers will follow once one has developed the required skill-sets and gained the experience. Hence, chip design is an extremely fulfilling career option for the technically and technologically inclined. As for the areas to look out for, the general consensus is that chips designed for consumer electronics is a growth area. Designs for wireless devices and multi-core processor design is bound to go up too, while integration of MEMS is a growth area that has great applications in the healthcare and medical arena. Entertainment and graphics technologies coupled with designing chips that give you higher performance at lower power consumption should be the focus for young entrants. "The dawn of energy age has increased demand on semiconductors to do more with less energy (performance per unit of energy). Another trend is towards always-on mobile connected devices. This is giving push to cloud computing where most of the horsepower is in a data centre and a smart connected device can access this capability from anywhere," says Dr. Atul Garg (PhD), a senior chip designer at NVIDIA Corporation, Santa Clara, USA. The engineer from IIT Kanpur, who now holds ten design patents to his credit has this final advice for aspirants: "Keep yourself current with state of the art in such a fast moving industry, increase productivity to be in sync with industry trends, and keeping ahead of the competition. Learning is a lifelong process and nothing is impossible if you work hard." ■

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